From Equity Pairs to Options Relative Value

Does learning improve capital allocation?

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Abstract

We extend equity-pairs trading to relative implied volatility in approximately 30-day, 25-delta puts on related equity underlyings. A screened opportunity rule generates trades; learning then determines their entry size. In a simulated $10 million account over January 2022–August 2025, constant full sizing achieves annualized excess Sharpe 1.055, compared with 1.263 for decision-tree states and cross-entropy sizing. The observed gain of 0.208 has a conditional 95% interval of $[-0.103, 0.424]$. The learner admits 110 trades compared with 75 under constant full sizing. Performance varies across years, and previously inspected history, different training and evaluation conventions, and simplified execution limit the conclusions. The comparison supports positive historical profit above cash and a higher point estimate for learned sizing, without establishing a repeatable advantage from learning.

1 Introduction

Suppose AMD and NVDA face similar news, but AMD options become unusually expensive relative to NVDA options. Selling the richer options and buying the cheaper ones expresses a view about the relative price of volatility. Daily stock hedges reduce immediate directional exposure, and initial vega matching reduces sensitivity to a common small change in implied volatility. The trade still carries risk: related stocks can experience different events, and option premiums may correctly reflect those differences.

Our starting point is Ning and Lee's equity-pairs framework [1], which separates constructing a spread from learning how to trade it. We carry that distinction into options. First, we ask whether a screened relative-volatility rule produces useful historical trades under explicit contract and funding assumptions. We then ask the central question: *does learning improve how capital is allocated to those opportunities?* A profitable signal and a useful allocation policy are separate findings.

After explaining the motivating paper and our adaptation, we define the trade, monthly training procedure and tree/CEM sizing method. We then compare constant and learned sizing in the same funded account and examine the learner's annual results. The comparison uses the retained configuration and saved historical backtests.

2 Motivating paper and adaptation

What the paper does

Ning and Lee [1] study how to construct and trade mean-reverting equity spreads. At time t, let $S_{1,t}$ and $S_{2,t}$ denote the two stock prices. They form the spread $X_t = S_{1,t} - BS_{2,t}$, where B is the hedge coefficient, and choose B by grid search to minimize empirical mean reversion time: the average duration from a significant local extreme to the next mean crossing. Formation does not fit an Ornstein–Uhlenbeck (OU) mean-reversion model.

For trading, they use tabular Q-learning. The state encodes recent spread moves by direction and magnitude; available actions buy, hold or sell a unit, subject to the current position. The policy enters a long spread and later closes it, and does not initiate a short spread. Its reward favors buying below and selling above the equilibrium mean, with a transaction-cost term $c|A|$, where c is the cost coefficient and $|A|$ is the magnitude of the trading action. Q-values combine immediate reward with the estimated value of later decisions.

Because the reward requires a true equilibrium mean, the agent trains on 10,000 one-year simulated OU paths before application to historical stocks. The market study forms ten US pairs in 2022 and trades them in 2023. Published daily Sharpe ratios exceed the Distance Method and OU benchmarks for all ten pairs. Those daily figures are not directly comparable with our annualized options Sharpe ratios.

Criticisms that motivate our design

Three issues limit interpretation. **Short history:** one trading year provides limited evidence across regimes. **Costs and scale:** the reward defines a cost term, but its numerical implementation and reconciliation to net performance are not clearly documented. This is not evidence that costs were omitted. Higher turnover raises total costs; larger or faster orders can also increase per-unit slippage and market impact. A constant coefficient does not establish scalability. **Simulation transfer:** formation is empirical, but policy training still uses specified mean-reverting dynamics and a known equilibrium. Real relationships can shift, so extensive synthetic experience does not establish market transfer.

What we borrow and what we change

We borrow the separation of opportunity construction from trading decisions, and the question of whether learning improves on a simple rule. We change the instrument from equity-price spreads to option-implied-volatility spreads, using actual contracts, initial vega matching and daily stock delta hedges. Completed historical outcomes replace a known synthetic equilibrium for training. We make simulated costs explicit and compare multiple trading years.

The next section defines relative-IV opportunities, option contracts and hedging through expiry. Our regression tree groups completed historical opportunities; cross-entropy search chooses an entry size for each group. This is supervised partitioning followed by sizing search, rather than sequential Q-learning. Comparing with constant sizing measures the additional historical value of learned allocation.

3 Our strategy and evaluation

3.1 From a volatility signal to a hedged trade

Implied volatility (IV) reproduces an option quote under a pricing model; realized volatility (RV) measures stock-price variation. RV inputs use trailing data available at the signal date, not future outcomes. We trade puts near 25 delta and 30 days to expiry (P25), a contract choice retained after exploratory research and fixed across all sizing policies. We do not claim that this segment is optimal. Appendix A gives selection and quote rules.

For stocks A and B, let $IV_{A,t}$ and $IV_{B,t}$ be their selected options' implied volatilities on signal date t. Their difference, $D_t = IV_{A,t} - IV_{B,t}$, measures which stock's option has higher quoted IV. We rank D_t within the 252-session rolling history ending on the signal date, requiring at least 63 observed differences. Above the 80th percentile, the rule sells A's relatively rich put and buys B's relatively cheap put; at or below the 20th percentile it reverses the direction. The percentile compares the pair with its own history, allowing for persistent IV differences; the corresponding IV boundary moves with the rolling sample.

Each month, we estimate beta of A's daily returns on B's and their correlation using the 126 sessions preceding the month's first session. We require beta between 0.85 and 1.15 and correlation of at least 0.5. We then check whether historical simulations for the same pair and proposed direction have positive average net profit per unit of initial vega. These are full-size, standalone hedged simulations on available candidate dates, calculated with the training costs. This check uses outcomes completed within the previous 91 and 365 calendar days, requiring at least 21 simulations in the shorter window and 63 in the longer window. If either count is too small or either average is nonpositive, we do not propose the trade. These simulations may overlap and need not have been taken by the funded account.

Only simulations completed strictly before the signal date contribute. For example, a trade still open on 15 June cannot supply its eventual expiry profit to that day's decision. The overlapping outcomes are not independent observations.

Contract quantities approximately match initial vega: the change in option value for a one-percentage-point increase in IV. Table 1 is an illustrative example of the initial positions, with each put at delta -0.25 per share and each contract representing 100 shares.

Position	Contracts	Vega per IV point	Initial stock hedge
Buy A puts	+100	+\$450	Buy 2,500 shares
Sell B puts	−150	−\$450	Short 3,750 shares

Table 1: Initial vega matching. A contracts have \$4.50 vega each; B contracts have \$3.00.

The example matches local vega exposure, not the two stocks' eventual risks. We hold options to expiry and adjust stock delta hedges daily. Final profit combines entry premiums, settlement and hedges along the realized stock paths, after costs and financing. We evaluate this complete trade; interim IV convergence alone does not determine expiry profit.

3.2 Simulating trades in a funded portfolio

We evaluate each sizing policy in a single $10 million account. The policies receive the same trading opportunities and follow the same execution, hedging and account rules. This comparison measures how their sizing decisions affect net portfolio performance.

For each opportunity, we select the two put contracts—including their strikes and expiries—using information available on the signal date. We attempt to enter those same contracts on the next trading day. Quantities are rounded to whole contracts, with no more than a 10% difference between the initial long and short vega exposures. A proposed trade is accepted only if usable quotes are available and it satisfies the account's capital and risk limits. The common opportunity set comes from the retained historical formation calendar and pair panel listed in Appendix B.

Once accepted, the option positions are held to expiry and their stock delta hedges are adjusted daily, except for corporate-event exits described in Appendix A. All positions share the account's cash and available capital. Stock hedges in the same company are combined into a net position. Consequently, an existing trade can reduce the account's ability to accept another opportunity.

Portfolio performance includes option premiums and settlement payoffs, stock-hedge gains and losses, dividends, transaction costs and financing. Option trades are recorded at the quoted midpoint, with an execution charge equivalent to 0.10 IV percentage point of traded vega and a commission of $0.03 per contract. Stock-hedge transactions cost one basis point of traded value. Cash earns the benchmark interest rate; negative cash pays that rate plus 1% annually, and short stock incurs a 0.3% annual borrowing charge. Appendix A gives the cost formula and account limits.

Because unused cash earns interest, we measure the strategy's contribution relative to an account invested entirely at the benchmark cash rate. This separates the value added by trading from interest earned on the account balance.

3.3 Training, selection and testing

Each month, we train a new tree and size map using historical examples from the preceding 18 months, then apply that model to the next month's entry decisions. Figure 1 separates this rolling process from the choice of model specification. Training and application use successive slices of the same market history.

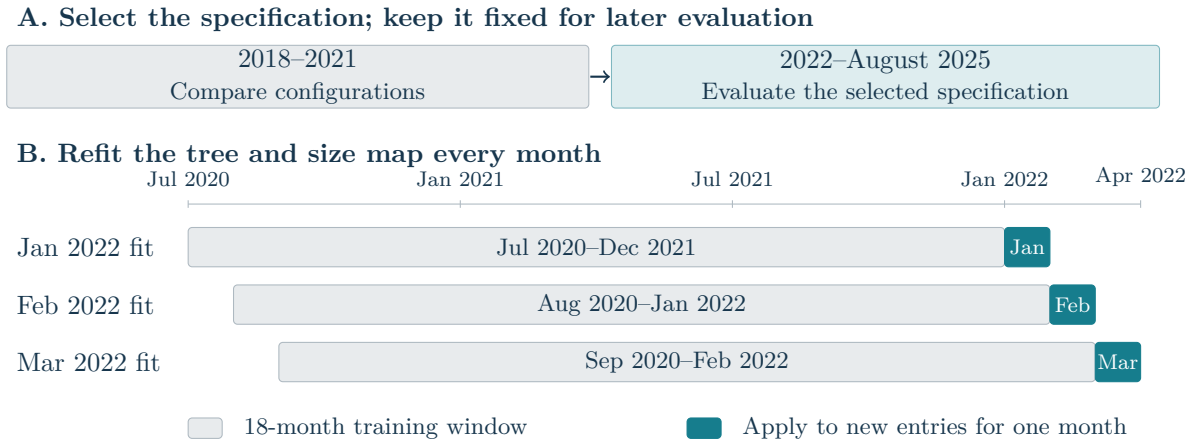


Figure 1: Two decisions, one historical dataset. The upper panel selects the 27-input, 18-month, 50/75/100% specification using earlier performance. The lower panel illustrates monthly refitting during evaluation. A training example is included only if its signal date falls within the window and its trade outcome is completed before that month's fit.

We retain the configuration with the highest annualized daily excess Sharpe on 2018–2021 among 33 existing configurations spanning training-history, feature and sizing-action choices. The selected configuration uses 27 inputs, an 18-month window and 50/75/100% actions. Candidate identifiers break score ties; its selection score is 0.575 in a $345 million research account. This configuration is then replayed at $10 million; the account assumptions are summarized in Appendix A.

Each monthly fit uses candidates that passed the entry screens on their signal dates, have valid simulated outcomes, and finished before fitting. These examples need not have been accepted by the funded account. Only signal dates within the preceding 18 months enter the fit. Tree fitting and CEM use this same sample; the wider candidate history supplies the profitability screens.

The later comparison runs from 3 January 2022 to 29 August 2025. The model specification stays fixed while monthly refitting continues: a completed March trade can inform an April fit, but an unfinished outcome cannot. New signals stop on 30 June 2025. All accounts continue from 2018 with their own accumulated cash and holdings; the 2022 boundary is a reporting split, not an account reset.

Later history had already been examined during strategy development. Monthly fits respect time order, but this period is not independent of earlier choices of contracts, screens and universe. Training also uses $0.65 commission per option contract, compared with $0.03 for every policy in final evaluation. We therefore interpret the comparison as historical evidence.

We report annualized Sharpe of daily returns above the cash benchmark, profit above cash and cash-relative drawdown. Appendix A.5 defines these metrics and the opening portfolios at the reporting boundary.

3.4 Learning an entry size: decision tree and CEM

The opportunity rule fixes the pair, direction and contracts. We learn only the entry size: 50%, 75% or 100% of a reference budget of $450 vega per IV point on each leg, scaled by current account equity relative to its initial equity. Opportunities differ in their historical outcomes, so the learner groups them and assigns a common size to each group (Figure 2).

Each monthly regression tree receives 27 inputs: 15 numerical characteristics and 12 availability flags, covering relative IV, trailing realized variance, relatedness, maturity, the CBOE Volatility Index (VIX), costs and completed profitability. Appendix A.4 defines the inputs. It learns splits using historical full-size profit above cash as its target, splitting to reduce squared prediction error [3]. Each of its up to eight leaves is a state. The cross-entropy method (CEM) [2] then chooses one size for each state.

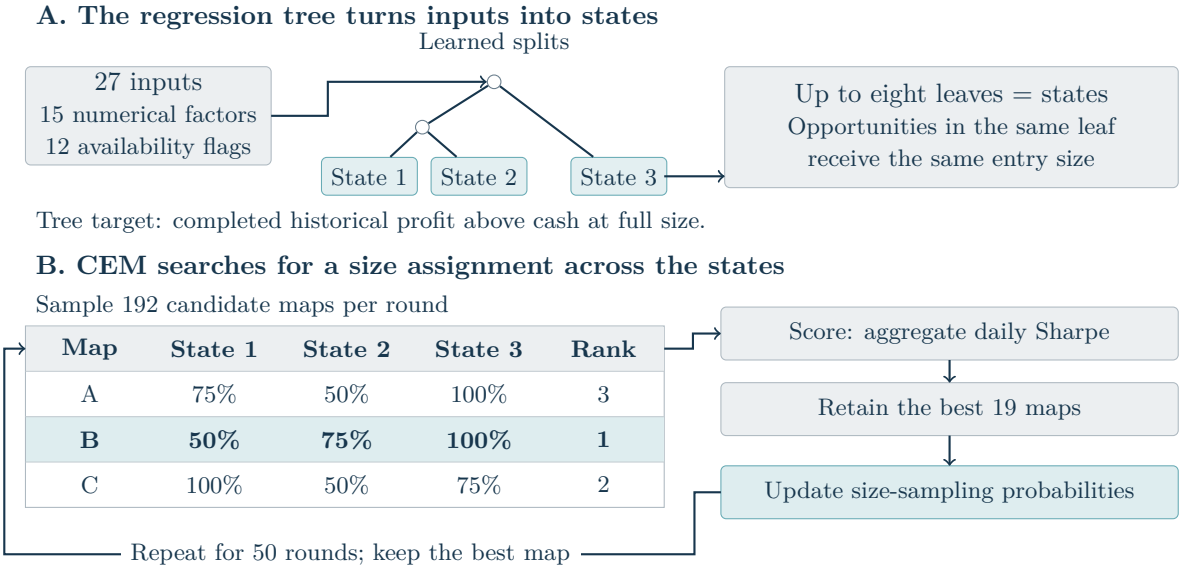


Figure 2: From market inputs to entry sizes. The three-state tree, candidate assignments and ranks are illustrative; actual monthly trees have up to eight states. CEM searches over complete maps, with one 50/75/100% size per state. The selected map is used for the next month's new opportunities.

How CEM scores a size map

A candidate map $\boldsymbol{m}$ assigns an entry size a_ℓ—50%, 75% or 100% of the reference budget—to each tree state ℓ. For training example i, let x_i denote its 27 inputs and $\ell(x_i)$ the fitted tree's state. Let $p_{i,d}(a)$ be its net profit above cash on day d at size a. Each size has its own rounded contract quantities and standalone cash and hedge flows; smaller-size outcomes are not proportional rescalings of the full-size outcome. Stored paths share a fixed normalization, which cancels from the Sharpe score.

We sum the daily profits across the completed training examples:

$$g_d(\boldsymbol{m}) = \sum_i p_{i,d}\Big(a_{\ell(x_i)}\Big)\,.$$

Here $g_d(\boldsymbol{m})$ is combined profit above cash on day d. We rank maps using

$$\widehat{S}_{\text{train}}(\boldsymbol{m}) = \sqrt{252}\,\frac{\overline{g(\boldsymbol{m})}}{s_{g(\boldsymbol{m})}}.$$

The numerator is mean daily profit over the sample's trading calendar; the denominator is its sample standard deviation. The factor $\sqrt{252}$ annualizes using 252 trading days per year. Overlapping trades and zero-profit sessions remain in the calculation. This is a training score based on standalone profit paths; the final account Sharpe uses percentage returns and applies joint admission constraints.

CEM repeatedly samples maps, ranks them and updates the probability of each size in each state toward the best 19 of 192 maps. Each round also includes the three constant-size maps and the best map found so far in that monthly search. After 50 rounds we retain the best-scoring map, without claiming a global optimum. For each new qualified proposal, the tree assigns its state, the map requests its size, and the funded account accepts or rejects it.

4 Benchmark and learned-sizing results

4.1 Does learned entry sizing improve the benchmark?

We compare the tree/CEM policy with constant requests of 50%, 75% and 100% of the same reference budget. Each policy receives the same proposals and keeps the proposed direction and contracts.

Figure 3 shows when profit above cash accumulates. Most of the learner's final lead develops during 2022; both policies lose ground during 2024.

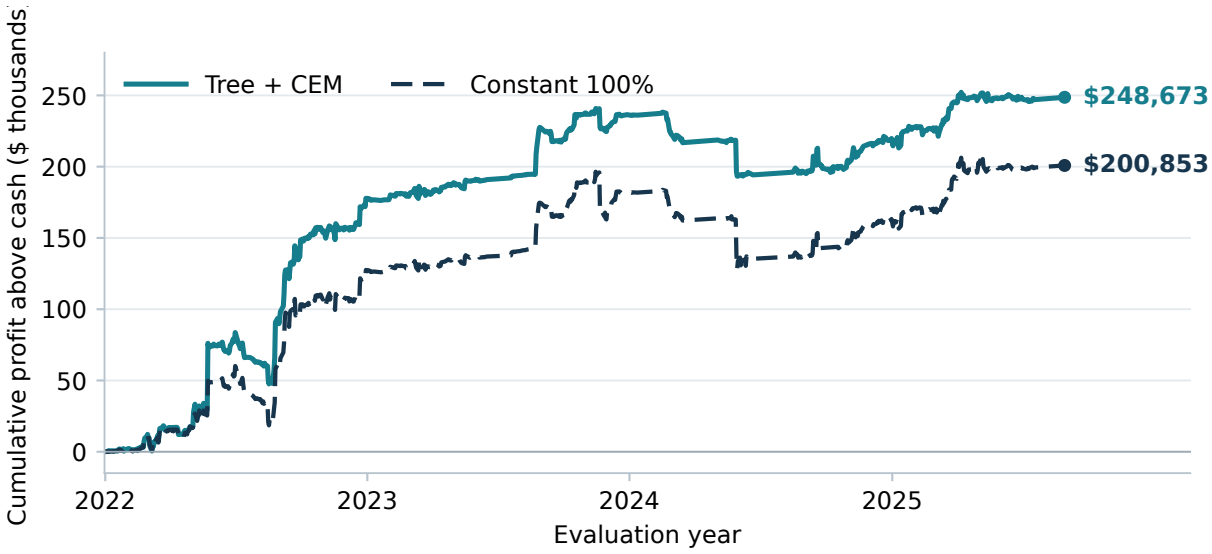


Figure 3: Cumulative profit above cash, 3 January 2022–29 August 2025. Curves are rebased at the 31 December 2021 close; the books retain carried positions. Endpoints match Table 2. The paired 95% interval for the Sharpe gain, $[-0.103, 0.424]$, includes zero.

Table 2 compares the learner and constant sizing with identical proposals and final funded-account rules. Constant full sizing provides a useful historical benchmark: later excess Sharpe is 1.055 and profit above cash is $200,853. Learned sizing raises those estimates to 1.263 and $248,673, an observed Sharpe gain of 0.208 and approximately $47,800 additional profit above cash.

Policy	Earlier Sharpe	Later Sharpe	Later entries	Profit above cash ($)	Relative drawdown (%)
Constant 50%	0.437	1.007	151	204,913	0.735
Constant 75%	0.449	0.944	104	195,021	0.748
Constant 100%	0.478	1.055	75	200,853	0.646
Tree + CEM	0.573	1.263	110	248,673	0.495

Table 2: Earlier: 2018–2021 replay at $10 million; later: January 2022–August 2025. Entries, profit and maximum cash-relative drawdown refer to the later period. Earlier scores are $10 million replay scores, not the $345 million configuration-selection scores.

The paired circular-block bootstrap, using 126-session blocks and 1,000 replications, gives a 95% interval of $[-0.103, 0.424]$ for the Sharpe gain. It includes zero and omits search adjustment. The observed gain therefore does not establish a repeatable sizing improvement. The learner admits 110 trades against 75. The comparison measures the combined effect of requested sizes and the portfolios admitted under the account rules; it does not isolate gains from sizing trades common to both policies.

The account earns annualized total return of 4.667%, while annualized growth relative to an all-cash account is 0.530%. Most total account growth therefore comes from cash interest. Trading is active—110 new entries—but its net contribution above cash is modest: \$248,673 over the later period. Average reserve is 2.44% of equity and peak reserve 15.77%; stress and pair limits can reject entries even with a low account reserve. We use excess Sharpe and profit above cash to assess the strategy's added value, rather than attributing the full 4.667% account return to trading.

4.2 How does performance vary across years?

Table 3 breaks the learner's later performance into calendar periods. Its strongest annual cash-relative performance occurs in 2022, which also accounts for 66 of the 110 later entries. In 2024, the account earns a positive total return but loses relative to cash. The partial 2025 period is positive above cash again. Together with Figure 3, these results show that the aggregate profit reflects uneven performance across years.

Year	Account return (%)	Above cash (pp)	Excess Sharpe	Entries
2022	3.470	1.673	2.458	66
2023	5.288	0.432	1.492	19
2024	5.104	−0.283	−0.907	11
2025 (partial)	3.207	0.169	1.092	14

Table 3: Tree/CEM annual funded results; 2025 ends on 29 August. Account and above-cash returns are unannualized for each displayed period, including partial 2025. Above-cash figures are return differences in percentage points; Sharpe is annualized.

5 Conclusion and limitations

We compare constant sizing with a tree/CEM policy that learns entry sizes from completed historical trades. Both policies use the same screened options opportunities, hedging rules and funded-account constraints.

The constant-full rule earns positive historical profit above cash. Learned sizing raises the funded point estimate, with a 0.208 Sharpe gain and approximately \$47,800 additional profit, but its gain interval includes zero. It also admits more trades, 110 versus 75, under the same account rules. Annual results vary, including a loss relative to cash in 2024. The evidence supports a higher aggregate historical estimate for learned sizing, with uncertainty about its repeatability.

The comparison uses previously inspected history and differs across training, selection and evaluation in capital, costs and admission. Execution and option exercise are simplified. These results establish neither repeatable learning benefits nor live profitability or capacity. Most account growth is cash interest; trading value is measured above cash.

A Implementation details

A.1 Opportunity specification

The retained segment uses 30-day expiry anchors from the historical contract selection, then chooses a put in that expiry with absolute delta closest to 0.25, within 0.075. Ties favor narrower relative spreads, higher open interest and lower contract identifier. Quotes must be standard 100-share contracts, with positive bids, noncrossed markets, open interest of at least 100 and bid–ask width at most 25% of midpoint. The 30-day expiry anchor favors eligible maturities closest to 30 days; next-session entry additionally requires 7–55 days remaining. Identifiers are fixed on the signal date.

Relatedness uses 126-session daily simple-return covariance divided by B's return variance for beta of A on B, and the corresponding correlation. Values are calculated before the first session of each month and held fixed that month. The IV percentile uses available differences in a 252-session window including the signal close, with a minimum of 63 observations. Profitability screens use completed full-size simulations in each direction, including available dates that did not qualify as funded proposals. Their net dollar profits are divided by mean initial leg vega before averaging; completion dates determine the 91- and 365-calendar-day windows.

A.2 Risk, execution and accounting conventions

The account checks the proposed portfolio after each entry. Stress loss is the sum of each underlying's worst nonnegative modeled loss across nine combinations of spot shocks of $-20/0/+20\%$ and IV shocks of $-10/0/+10$ percentage points, including its delta hedge. It takes the worst shock for each name separately, without credit for pair correlation. Stress loss may not exceed 5% of the applicable equity budget.

Reserve is gross marked option premium plus half the stock-hedge notional summed in absolute value across names after netting within each name. Reserve may not exceed 80% of the equity budget, stock-hedge notional may not exceed twice that budget, and gross vega is capped at \$20,000 per IV point per \$1 million budget. Global checks use account equity; each pair's checks use \$1 million scaled by current account equity divided by initial account equity. The additional contract-segment check has the same budget and is redundant with the pair check in this one-segment study. Same-date proposals are processed in stable pair-identifier order. These are research admission limits, not broker margin or guarantees against later loss.

The simulation cash-settles expiry and supports verified integer splits. Announced corporate actions or unsupported declared distributions trigger exits of both legs of affected pairs. Missing entry quotes reject the proposal; missing held quotes can be marked using last-known IV with current spot, rates and known dividends. Black–Scholes valuation approximates European exercise for American equity options. Market impact, early assignment and forced liquidation are not fully modeled.

We book option trades at quoted midpoint and deduct a cash execution penalty equal to 0.10 IV percentage point times the traded vega. For day t, the evaluation costs are

$$C_t^{\text{execution}} = \sum_{j \in \mathcal{F}_t} |\Delta N_{j,t}| \big(0.10\, V_{j,t} + 0.03\big) + 0.0001 \sum_s P_{s,t} |\Delta h_{s,t}|, \tag{1}$$

$$C_t^{\text{funding}} = \big[0.003\, H_{t-1}^{-} + 0.01 \max(-B_{t-1}, 0)\big]\, \Delta\tau_t.$$

The following notation identifies the quantities in the cost calculation.

Symbol	Meaning and unit
t, j, s	Trading day, option-contract index and stock index, respectively.
$C_t^{\text{execution}}$, C_t^{funding}	Execution and funding charges on day t, in dollars.
$\mathcal{F}_t$	Set of option fills charged execution costs on day t.
$\Delta N_{j,t}$	Signed change in the number of contracts of option j; absolute value counts contracts traded.
$V_{j,t}$	Dollar vega of one contract per IV percentage point (100 times per-share quote vega).
$P_{s,t}$	Stock s's transaction price, in dollars per share.
$\Delta h_{s,t}$	Signed change in the net stock hedge, in shares; $P_{s,t}\lvert\Delta h_{s,t}\rvert$ is dollar turnover.
H_{t-1}^{-}	Previous day's short-stock notional, in dollars.
B_{t-1}	Previous day's cash balance, in dollars; a negative balance represents borrowing.
$\Delta\tau_t$	Elapsed calendar days divided by 365, in years.

The coefficients charge 0.10 IV percentage point of option vega, \$0.03 commission per contract and one basis point (0.0001) of stock turnover. Funding charges use 0.3% (0.003) annual short-stock borrow and a 1% (0.01) surcharge on negative cash.

Benchmark-rate cash interest is recorded separately as $I_t = r_{t-1}B_{t-1}\Delta\tau_t$, where I_t is interest in dollars and r_{t-1} is the previous day's annual benchmark rate, expressed as a decimal. Negative cash therefore pays the benchmark rate as well as the surcharge above. Expiry settlement carries no option commission or execution penalty.

A.3 Tree states and cross-entropy search

Stage	Capital, costs and objective
Monthly fitting	Standalone \$1 million trade paths, \$0.65 option commission; tree predicts full-size outcome, CEM maximizes daily profit Sharpe without joint portfolio admission.
Configuration selection	\$345 million shared account, \$0.03 commission; highest 2018–2021 excess-return Sharpe among 10 history/recency, 11 feature and 12 action/history configurations. Selected score: 0.575.
Final comparison	\$10 million shared account, \$0.03 commission; same proposals, execution and admission rules for every policy.

All stages use a 0.10-IV-point option execution penalty and one basis point of stock-hedge turnover. Selection and evaluation use shared accounts with \$1 million pair budgets scaled by equity; fitting uses standalone paths. These differences are retained limitations of the existing study.

The tree targets cumulative full-size profit above cash, with at most eight leaves. For n completed examples, minimum leaf size is $\max(8, \min(63, \lfloor n/8 \rfloor))$. Here $\lfloor n/8 \rfloor$ rounds $n/8$ down to an integer; the rule sets the minimum examples per leaf to one eighth of the training count, bounded between 8 and 63. Fewer than 16 examples trigger full-size fallback. Every available action requests nonzero exposure, although account admission can reject it. Each month has its own tree and action map; split features, boundaries and size assignments can change while the 27-input specification remains fixed.

CEM starts each month with uniform probabilities. Its update is 0.97 times a mixture of 30% current probability and 70% frequency among the best 19 maps, plus 0.03 divided equally across the three actions. The seed is 2026.

A.4 Inputs available at the signal date

The model uses the following 15 numerical inputs. Proposed-direction quantities are oriented as long leg minus short leg. IV and volatility levels are decimal annualized values. Trailing realized variance is 252 times mean squared daily simple returns over the latest 20 listed trading sessions, with at least 10 observations, including the signal-close return. Returns use the retained historical return series (generally excluding dividends), with adjusted-close percentage changes where that series is unavailable.

Input	Definition
Mean IV	Mean of the two selected-contract IVs.
Oriented IV gap	Long-minus-short IV divided by mean IV.
Rank strength	Twice the absolute distance of the IV-gap percentile from 0.5.
Oriented variance-premium gap	Long-minus-short difference in IV squared minus trailing realized variance, divided by mean IV squared.
Mean RV	Mean of the square roots of the two trailing realized variances.
Beta; correlation	Two inputs: 126-session A-on-B beta and correlation, fixed at monthly formation.
Mean expiry; expiry gap	Two inputs: mean calendar days to expiry and long-minus-short expiry days.
Commission in IV points	Sum across legs of \$0.65 divided by each contract's dollar vega per IV point.
VIX	Most recent strictly prior VIX close divided by 100.
91-day profit; count	Two inputs: completed full-size profit per initial vega averaged over the trailing 91 calendar days and the number of completed outcomes. The profit input is divided by 100 as a fixed scale.
365-day profit; count	Two inputs: the same measurements over 365 calendar days.

The first three inputs are core fields. Each of the other 12 has a binary availability flag, giving 27 inputs in total. Nonfinite optional values receive zero and flag zero; observed values have flag one. Optional missingness is not an additional entry veto. Profit features use only outcomes completed strictly before the signal date. Future stock variation and eventual trade profit enter outcomes, not decision inputs.

A.5 Performance measurement and the reporting boundary

Let N_t be account equity and K_t all-cash benchmark wealth at close t. Both start at \$10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over elapsed calendar days using a 365-day year. Daily excess return is

$$e_t = \left(\frac{N_t}{N_{t-1}} - 1\right) - \left(\frac{K_t}{K_{t-1}} - 1\right).$$

Annualized excess Sharpe is $\widehat{S} = \sqrt{252}\,\bar{e}/s_e$, where $\bar{e}$ and s_e are the sample mean and standard deviation of daily excess returns, and 252 is the assumed trading days per year.

For a reporting period with preceding close 0 and final close T, dollar profit above cash is $(N_T - K_T) - (N_0 - K_0)$. Relative growth uses the ratio $Q_t = N_t/K_t$: annualized growth relative to cash is $(Q_T/Q_0)^{1/Y} - 1$, where Y is elapsed calendar days divided by 365.25. Cash-relative drawdown is the largest percentage decline in Q_t from its running peak within the reporting period, including close 0. Reported reserve percentages are reserve divided by current equity.

Annual-table account returns are $N_T/N_0 - 1$; above-cash percentage points subtract the corresponding benchmark return $K_T/K_0 - 1$. These period returns are unannualized, including partial 2025. Sharpe remains annualized.

The later period starts after the 31 December 2021 close. Each policy retains its own earlier gains, cash and open trades; benchmark wealth is \$10.510 million for all policies at that close. Figure 3 subtracts each policy's own opening dollar excess wealth, so curves begin at zero without resetting the underlying accounts.

Policy	Equity (\$m)	Cash (\$m)	Open trades	Hedge notional (\$m)
Constant 50%	10.553	10.581	5	0.190
Constant 75%	10.564	10.592	4	0.198
Constant 100%	10.581	10.624	3	0.235
Tree + CEM	10.568	10.594	4	0.190

Hedge notional is summed in absolute value after netting within each name. This is a comparison of continuing policies with different opening portfolios, not a common-inventory experiment started in 2022.

B Pair universe and date coverage

Historical option quotes, underlying prices, rates and dividends come through WRDS. We reuse the 345-pair roster retained in earlier project research; its membership and historical formation calendars are inherited rather than selected anew for this comparison. The roster includes equities and ETF underlyings, including XOP/COP. Results are conditional on these prior universe choices, not evidence of an independently selected universe.

Of that roster, 125 pairs and 161 underlying securities have recorded candidate rows in the prepared P25 panel, totaling 36,327 observations. Preparation requires observed selected-contract IV, relatedness eligibility and the retained historical formation calendar. This panel is therefore a data-and-calendar-conditioned subset, not 125 pairs chosen by the sizing learner. Some candidates fail the IV or profitability screens: 46 pairs generate the 608 qualified proposals during 2018–2025, of which the learned funded account admits 207 entries, including 110 in the later period.

Pair identity uses historical security identifiers, avoiding duplicate pairs with reversed ticker order. Displayed labels retain the earlier roster's ticker names and orientation; they are not a claim of today's ticker identities.

The earlier selection period is 2 January 2018–31 December 2021. The later test is 3 January 2022–29 August 2025, with new signals ending on 30 June 2025 and the remaining positions followed through the test cutoff. Historical candidate examples before 2018 support training and profitability screens.

Table 4 lists every pair in the prepared panel. Candidate dates are the first and last recorded signal-date examples in that panel, including examples that fail entry screens; they are not a claim of continuous quote coverage or a pair's executed trading period. The proposal count refers only to qualified signals on the 2018–June 2025 formation calendar. A zero count means the pair supplies historical candidates but no qualified funded request in that calendar. Labels follow the archived roster.

Table 4: Prepared pair panel, recorded candidate-date ranges and qualified proposal counts. Dates are YYYY-MM-DD.

Pair	Candidate dates	n	Pair	Candidate dates	n
A/TMO	2010-09-14–2025-06-11	12	HES/XOM	2011-03-01–2024-06-06	27
ACLS/LRCX	2018-01-30–2025-01-30	0	HIG/CB	2011-03-01–2025-01-29	0
ADM/BG	2011-09-06–2024-07-23	0	HIG/TRV	2011-06-13–2023-08-07	0
AEM/GG	2010-08-05–2018-10-24	0	HL/KGC	2010-09-01–2024-11-26	0
AEM/NEM	2010-08-05–2023-04-18	0	HWM/GLW	2011-10-04–2016-04-28	0
AEO/GES	2023-02-02–2023-05-19	0	JPM/C	2015-03-02–2022-04-28	4
AEO/GPS	2014-06-03–2018-02-22	0	JPM/WFC	2011-10-04–2023-01-31	22
AEO/TJX	2010-09-01–2022-09-28	9	KBH/LEN	2010-10-05–2025-01-30	2
AIG/CB	2010-11-04–2025-06-30	0	KBH/TOL	2011-05-05–2025-03-26	22
AIG/HIG	2015-02-02–2024-02-20	0	KGC/NEM	2010-11-02–2025-06-23	0
AIG/TRV	2010-11-04–2023-09-20	0	KKR/APO	2022-04-20–2025-06-23	3
AMD/MU	2019-04-01–2023-04-28	23	KSS/BURL	2020-05-06–2023-11-22	8
AMD/NVDA	2012-03-06–2022-12-30	6	KSU/UNP	2010-11-02–2021-11-23	0
AMTD/ETFC	2016-06-16–2019-11-22	0	LEN/TOL	2010-09-02–2023-04-25	0
ANDV/PSX	2013-05-07–2018-04-18	0	LH/DGX	2011-10-10–2017-04-12	0
ARKK/TSLA	2022-08-05–2022-10-31	0	LHX/NOC	2010-10-08–2020-07-27	1
ATI/STLD	2010-08-11–2021-03-05	0	LNG/KMI	2013-05-07–2025-06-18	0
ATVI/EA	2017-03-02–2021-09-23	0	LNG/OKE	2018-08-15–2022-08-25	0
AU/NEM	2011-11-01–2023-06-21	0	LNG/WMB	2013-07-02–2025-06-17	0
BA/RTX	2010-08-05–2023-02-16	0	LYFT/UBER	2020-05-05–2025-06-23	0
BAC/C	2011-01-04–2024-04-30	68	M/KSS	2010-08-05–2022-01-31	14
BAC/JPM	2010-08-05–2025-06-26	76	MAC/SPG	2010-08-16–2024-12-17	0

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Table 4 continued

Pair	Candidate dates	n	Pair	Candidate dates	n
BAC/WFC	2010-08-05–2025-06-26	14	MAR/HLT	2015-11-02–2025-06-24	5
BAX/MDT	2016-10-11–2024-04-19	0	MET/PRU	2010-08-05–2025-06-23	6
BLK/GS	2013-09-03–2023-03-30	3	MGM/WYNN	2010-10-05–2024-09-25	2
BLK/JEF	2022-12-07–2022-12-29	0	MSFT/META	2020-04-01–2020-07-31	1
BTU/AMR	2022-04-11–2025-05-19	0	MSFT/V	2018-04-03–2023-05-31	2
BTU/CEIX	2022-06-16–2025-06-11	0	MU/NVDA	2011-11-03–2020-05-22	0
BTU/HCC	2022-03-23–2025-06-11	0	NEE/FE	2023-11-15–2025-02-20	0
BZH/DHI	2015-02-06–2025-05-27	0	NOC/RTN	2010-08-17–2019-04-25	3
BZH/KBH	2014-10-15–2025-05-19	0	NVTA/NVCR	2020-05-11–2020-06-23	0
BZH/TOL	2013-11-05–2025-05-27	0	OIH/HAL	2015-06-12–2025-05-27	34
C/WFC	2013-01-04–2025-06-26	32	OIH/SLB	2012-08-07–2025-06-25	32
CAL/SKX	2021-04-08–2022-10-27	0	OIH/XOP	2016-11-09–2025-06-25	0
CAR/HTZ	2020-04-01–2020-04-20	0	OKE/KMI	2015-04-21–2025-06-18	0
CCL/NCLH	2020-01-22–2025-06-30	3	OKE/WMB	2015-02-13–2025-06-17	0
CE/HUN	2022-10-25–2025-04-24	0	PENN/CZR	2020-04-07–2025-06-23	2
CENX/AA	2017-06-22–2025-06-16	0	PENN/WYNN	2021-07-02–2022-09-28	0
CHK/SWN	2011-09-06–2019-06-17	0	PLUG/BE	2021-05-14–2024-03-27	0
CLF/FCX	2010-08-05–2022-08-31	0	RCL/NCLH	2016-07-19–2023-02-16	2
CLF/TECK	2010-08-06–2025-06-17	0	RIG/GLNG	2019-02-01–2022-11-21	0
COR/MCK	2017-12-18–2022-10-24	0	RRC/SLB	2011-09-09–2022-01-20	0
CSX/UNP	2010-08-06–2025-05-19	0	SGMS/IGT	2018-09-04–2020-06-25	0
CVX/XOM	2010-08-05–2025-06-26	3	SNAP/UBER	2022-04-14–2022-11-30	7
CYH/THC	2014-09-05–2016-02-12	0	SPWR/CSIQ	2017-04-20–2024-02-21	0
DHI/KBH	2022-08-01–2023-03-29	1	T/VZ	2012-01-03–2023-10-31	1
DHI/TOL	2010-08-10–2023-04-20	2	TER/QRVO	2020-07-06–2023-02-17	1
DHR/TMO	2022-04-04–2025-06-18	23	THC/HCA	2013-07-05–2025-06-30	14
DUK/SO	2010-10-05–2023-09-21	0	THC/UHS	2015-06-01–2015-10-23	0
ELV/CNC	2020-05-05–2020-07-28	0	TLRY/ACB	2021-04-16–2022-04-05	0
ETSY/Z	2022-09-06–2022-09-29	0	TRGP/KMI	2016-09-06–2025-05-22	0
FCX/TECK	2015-09-08–2025-06-17	0	TRV/CB	2011-11-04–2025-06-12	3
FE/SO	2010-11-03–2024-03-15	0	TXT/UTX	2011-05-03–2023-02-16	0
FNSR/VIAV	2017-09-19–2017-12-15	0	UAL/AAL	2014-09-12–2025-06-30	31
GE/RTX	2010-08-05–2022-02-22	2	UAL/DAL	2010-08-05–2025-06-30	36
GES/TJX	2012-03-06–2023-04-10	0	UHS/HCA	2015-11-16–2025-06-20	0
GG/NEM	2010-08-05–2019-01-17	0	UNH/ELV	2011-07-05–2025-06-30	0
GM/FCAU	2023-02-06–2024-02-21	0	UPST/COIN	2022-03-04–2024-02-29	3
GPS/GES	2022-10-07–2023-08-24	0	URI/AER	2018-03-05–2025-05-27	0
GPS/TJX	2010-09-01–2023-01-19	15	WMB/KMI	2015-06-12–2025-06-17	0
HAL/SLB	2010-08-05–2025-06-25	6	WYNN/HLT	2019-02-08–2022-12-28	0
HD/LOW	2016-02-04–2025-06-25	5	XOP/COP	2010-08-09–2025-06-26	17
HES/CVX	2011-02-01–2025-01-16	0			

C Reproducibility and evidence sources

The reported results use records retained in the repository's main-branch snapshot 688bc39. Numerical claims use the following saved records.

- Main sizing and annual results: `reports/p25_validation_split/periods.csv` and `annual.csv`. The main gain interval is in `holdout_intervals.csv`, and selection records are in `selection.json` in the same directory. Figure 3 uses the selected daily account in `data/p25_validation_split_experiment.zip` and the constant-full daily account under `data/submission/artifacts/contracts/p25_capital_sensitivity/replays/10m_fixed100`.
- Universe and model inputs: `reports/atm_nn_cost_hurdles/matched_roster.csv`, `data/submission/artifacts/contracts/p25_feature_search/training_rows.parquet` and `reports/p25_memory_ablation/formation_calendar.csv`. Feature definitions and fitting are in `scripts/uniform_smile_policy.py`, `scripts/same_structure_smile.py` and `scripts/run_p25_feature_search.py`; size-specific paths are in `scripts/curve_trading/unit_paths.py`.

The interval resamples the learner's and constant-full account's daily excess returns using the same day indices. For each of 1,000 replications, 126-session blocks start at randomly selected days, wrap circularly at the series end and are concatenated then truncated to the original length. We recalculate the Sharpe difference and use its 2.5th and 97.5th percentiles. Policies are fixed and accumulated research selection is unadjusted.

Vendor data retain their institutional license restrictions. Repository: https://github.com/kreitzz/rl-atm-vega-pairs.

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